

ProcessTensors.jl: An ITensor-native framework for non-Markovian open quantum dynamics

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Abstract

DRAFT: Non-Markovian quantum dynamics cannot, in general, be specified by a sequence of single-time reduced density matrices alone: multi-time statistics depend on how a system is interrogated and controlled between observations. Process tensors provide an operational description of this temporal structure and admit compact matrix-product-operator representations when temporal correlations are compressible. We present ProcessTensors.jl, an ITensor-native framework in which process tensors are Matrix Product Operators that can be constructed, stored, contracted with physical instruments, and reused for reduced dynamics and multi-time observables. The package extends the ITensors.jl/ITensorMPS.jl ecosystem with explicit Hilbert- and Liouville-space types, system and environment abstractions, instrument schedules, and high-level process-evaluation routines. Small environments can be treated with a dense builder that serves as a reference implementation, while an automated-compression-of-environments (ACE) backend provides a scalable route to multimode environments. We describe the mathematical conventions, software architecture, validation strategy, and representative workflows, and outline the package as an extensible platform for future process-tensor algorithms and characterisation tools. [TODO: insert one quantitative headline benchmark result before submission.]

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License information to appear upon publication.

Publication information to appear upon publication.

Received Date

Accepted Date

Published Date

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1 Introduction

The stage is yours. Write your article here. The bulk of the paper should be clearly divided into sections with short descriptive titles, including an introduction and a conclusion.

1.1 Existing software and the role of `ProcessTensors.jl`

1.2 Organisation of this paper

2 Liouville-space foundations

2.1 Vectorisation of operators

Use sections to structure your article's presentation.

Equations should be centered; multi-line equations should be aligned.

$$H = \sum_{j=1}^N \left[J(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z) - h S_j^z \right]. \quad (1)$$

In the list of references, cited papers [1] should include authors, title, journal reference (journal name, volume number (in bold), start page) and most importantly a DOI link. For a preprint [2], please include authors, title (please ensure proper capitalization) and arXiv link. If you use BiBTeX with our style file, the right format will be automatically implemented.

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2.2 Superoperators, Liouvillians, and channels

2.3 From spatial to temporal tensor networks

3 Software architecture

There is no strict length limitation, but the authors are strongly encouraged to keep contents to the strict minimum necessary for peers to reproduce the research described in the paper.

3.1 ITensor as the tensor-network substrate

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3.2 Hilbert- and Liouville-space objects

Figures should only occupy the strictly necessary space, in any case individually fitting on a single page. Each figure item should be appropriately labeled and accompanied by a descriptive caption. **SciPost does not accept creative or promotional figures or artist's impressions;** on the other hand, technical drawings and scientifically accurate representations are encouraged.

3.3 Systems and environments

3.4 A backend-oriented process-tensor API

4 Process tensors as first-class tensor-network objects

4.1 From a channel to a process

4.2 Anatomy of `ProcessTensor`

4.3 Building a small process tensor

4.4 Construction backends: dense reference and ACE

5 Quantum instruments

5.1 Operational meaning

5.2 Instrument organisation in `ProcessTensors.jl`

5.3 `InstrumentSeq`: a temporal analogue of `OpSum`

6 Contracting and evaluating processes

6.1 General contraction with `evaluate_process`

6.2 Reduced dynamics with `evolve`

6.3 Open process legs

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7 Common process-tensor workflows

7.1 Final reduced density matrices

7.2 Observable expectation values

7.3 Probabilities and trace-non-preserving branches

7.4 Multi-time correlation functions

8 Testers and enlarged interventions

Status: planned for the paper release; API under development.

9 Characterising process tensors and quantum memory

Section lead: Aaron Sander. The text below is a structural placeholder and should not be read as claiming currently released package functionality.

10 Validation and ACE benchmarks

10.1 Correctness against dense and exact references

10.2 Compression accuracy and temporal bond dimension

10.3 Runtime and memory scaling

10.4 Comparison with the C++ ACE toolkit

11 Conclusion and outlook

You must include a conclusion.

Acknowledgements

Acknowledgements should follow immediately after the conclusion.

Author contributions This is optional. If desired, contributions should be succinctly described in a single short paragraph, using author initials.

Funding information Authors are required to provide funding information, including relevant agencies and grant numbers with linked author's initials. Correctly-provided data will be linked to funders listed in the [Fundref registry](#).

Code and data availability The development repository is available at <https://github.com/Gauthameshwar/ProcessTensors.jl> and the documentation at <https://gauthameshwar.github.io/ProcessTensors.jl/stable/>. The final paper will cite an immutable release and archive the scripts/data used for all benchmark figures. [TODO: insert release DOI/Zenodo or SciPost codebase DOI at submission.]

A Broader Hilbert- and Liouville-space functionality

Add material which is better left outside the main text in a series of Appendices labeled by capital letters.

B Vectorisation conventions and index orientation

References

- [1] H. A. Bethe, *Zur Theorie der Metalle. i. Eigenwerte und Eigenfunktionen der linearen Atomkette*, Zeit. für Physik **71**, 205 (1931), doi:[10.1007/BF01341708](https://doi.org/10.1007/BF01341708).
- [2] P. Ginsparg, *It was twenty years ago today ...*, <http://arxiv.org/abs/1108.2700>.